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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Fast as Fahhhh
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8
Project / Solution Name	QuoteFlow
Name (Leader)	Lim Xin Jie
Email & Phone (Leader)	limxj0208@gmail.com & 011-35396866
University Affiliation(s)	EDUVO Academy
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian micro and small businesses often handle customer enquiries through messaging applications while preparing quotations separately. Staff must interpret requests, copy customer and item details, calculate totals, check quotation validity, prepare a document, and track whether a reply has been sent. For a small team, these repeated steps compete with sales and service work and create opportunities for missing information, inconsistent documents, and delayed responses.

The prototype supports all the business and for demo we use three fictional business profiles: Printing, Cleaning, and Repair. Primary users are staff preparing quotations and managers responsible for pricing and approval. Customers benefit from clearer clarification and more consistent documents. Telegram demonstrates the current journey; WhatsApp, Instagram, and Facebook are future integrations.

The economic issue is staff productivity: time spent transferring information between messages and documents reduces capacity for customer-facing work. We propose assisting this workflow while retaining human judgment over prices and outgoing commitments. The intended contribution is to SDG 8 through technology-enabled productivity and support for small-enterprise operations.

We will validate the size of this problem through observed manual quotation tasks. Time savings, reduced errors, and improved customer outcomes have not yet been measured.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

QuoteFlow connects customer enquiries with stored business and client context. Telegram is the implemented demonstration channel. AI extracts stated requirements, suggests draft items and clarification questions, and flags missing information. Unknown facts remain visible and prices stay blank. Staff reviews messages, asks clarification questions, apply selected suggestions, and prepares the quotation.

Staff explicitly submit a complete draft for Manager review. Manager enters prices, saves, and reviews the exact saved PDF and matches Telegram summary. The backend calculates decimal totals and expiry. Only explicit Manager approval authorizes delivery to the original Telegram chat. Version checks invalidate stale reviews. Activity and delivery outcomes remain visible; pending or unknown delivery blocks repeat sends, and partial success permits only the matching summary to be retried.

The application uses Python/FastAPI, PostgreSQL with SQLAlchemy/Psycopg, ReportLab, locally served PDF.js, and JavaScript ES modules/CSS with a Node web proxy. RunPod inference uses the hosted GLM-5.3 Public Endpoint through its OpenAI-compatible API and our private API key/account credits; we do not deploy a custom Pod or Serverless worker. Replit hosts the published demo. The team participated in workshops; completion of every workshop by every member is not claimed.

Local verification records 679 passing backend tests with 5 skips, 47 frontend tests, and 10 mocked actual-API checks, plus desktop/mobile browser checks. A separate synthetic RunPod request succeeded. Published inbox/settings availability was checked on 24 October 2026; the deployed version, full hosted Telegram journey and RP2 usage verification remain unverified. Manual drafting remains available when AI fails.

The proposed Orbyte fit is a reusable enquiry-to-quotation module using business context and role information. The prototype uses fictional records and simulated roles; live Orbyte integration and production identity are future work.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Layer	Technology and role
Application	Python 3.12+ and FastAPI; one backend for receipt, drafting, editing, PDF preview, approval, and activity.
Data	PostgreSQL with SQLAlchemy/Psycopg; shared enquiries, client/company context, quotation versions, PDFs, attempts, and activity.
AI	Runpod GLM-5.3 Chat Completions API; server-validated JSON extraction.
Documents	ReportLab; a fixed quotation layout using the saved version and decimal totals.
Messaging	Telegram Bot API; secret-checked webhook, Staff clarification replies and Manager-approved saved PDF/summary delivery to the original chat.
Interface and hosting	Implemented JavaScript ES modules/CSS UI, Node web proxy and locally served PDF.js. Replit demo is published; complete hosted acceptance remains pending.
Verification	pytest, pypdf and Node tests; isolated SQLite adapters and disposable PostgreSQL tests, mocked actual-API and desktop/mobile browser journeys. Report live-provider checks separately.

Architecture: Telegram webhook -> FastAPI -> PostgreSQL enquiry and business context -> RunPod extraction -> Staff preparation and submission -> Manager pricing -> saved-version ReportLab PDF and matching summary -> explicit Manager approval -> original Telegram chat and persisted activity.
The browser uses a Node web entry that proxies API requests to the localhost-bound Python backend.

Services used: Replit hosts the published demo. RunPod supplies the hosted GLM-5.3 Public Endpoint, called using our private API key and account credits.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our design centers on turning customer enquiries into reviewed quotations. Three fictional profiles demonstrate reuse across Printing, Cleaning, and Repair. Staff prepares requirements and explicitly submits them; Manager controls prices and offers outgoing approval. Unknown facts remain visible and manual drafting provides a fallback.

The frontend implements inbox, conversation, quotation editing, settings, and final PDF/summary review. Local browser journeys cover all three profiles and post-send Revision/Additional flows. SME interviews, novice-user walkthroughs, and paired preparation-time trials remain planned; no completed user-research or measured productivity result is claimed.

Replit hosts a Node web entry and private localhost-bound FastAPI backend connected to PostgreSQL and RunPod. Server-side secrets protect provider credentials. Published inbox and Business settings availability were checked. Fictional data and simulated Staff/Manager roles support a supervised demo; production authentication and live Orbyte integration are outside scope.

Next, verify the deployed version, hosted dependencies/security scan and complete team-controlled Telegram journey, including delivery recovery.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: SDG 8.2, improving productivity through technological upgrading and innovation. Secondary alignment: SDG 8.3, support for entrepreneurship and micro/small-enterprise growth.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Assume that preparing one quotation manually takes 15 minutes, including reading the enquiry, entering details, calculating totals, and creating a PDF. This assumption will be checked through timed trials.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage reduction in quotation preparation time. Compare manual preparation with QuoteFlow using the same fictional enquiries, measured from reading the enquiry to producing a correctly priced PDF ready for approval.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Target a 30% reduction in quotation preparation time compared with manual preparation, using the same fictional customer enquiries. All outgoing quotations must remain Manager-approved. This target has not yet been validated.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes [/] No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

[/] Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI; Pydantic; SQLAlchemy	Code libraries	MIT (installed package metadata)
Uvicorn; HTTPX	Code libraries	BSD-3-Clause (installed metadata)
Psycopg	Database adapter	LGPL-3.0-only (installed metadata); verify bundled binary notices
ReportLab	PDF code library	BSD license (installed metadata; retain license text)
PostgreSQL	Database software	PostgreSQL License; official source [2]
Tzdata	Time-zone data package	Apache-2.0 package metadata; retain included data notices

pytest; pypdf	Testing code libraries	MIT; BSD-3-Clause respectively (installed metadata)
GLM-5.3 by Z.ai via Runpod	Pretrained model/API	GLM-5.3 License [3]; Runpod service/event terms also apply
Telegram; Replit	Messaging API / published application host	Provider service terms; no open-source license assumed
Fictional enquiries and records	Team-created data	No external training dataset claimed; no model training planned

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes [/] No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

xinjie

25/10/2026